GS EXTERIOR EXPERTS · DECK DIVISION · PERMIT SET

# 281 S Jefferson Ave · Decks + Roof Covers

281 S Jefferson Ave, Louisville · two decks (12' x 12' + 14'8" x 3'8" walkway) on CEMCO steel and Diamond Piers · TimberTech Landmark American Walnut, picture frame + dividers, no butt joints · two lean-to covers, black standing seam, Hardie soffit / fascia / rakes · blinds kept · October 16, 2026

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Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale. This set carries no pricing.

# G-001 GENERAL NOTES · DESIGN CRITERIA

Loads, codes, structure summary, sheet index, site — the permit-set cover

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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## PROJECT

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel, TimberTech Landmark American Walnut)
281 S Jefferson Ave, Louisville CO 80027
197.8 SF · 1'-3" above grade · 2 zones

## DESIGN CRITERIA

| | |
|---|---|
| **Deck live load** | 40 psf uniform (IRC Table R301.5); guards 200 lb concentrated / 50 plf |
| **Ground snow** | 30 psf (ground; deck design uses the larger of snow and live) — 40 psf live governs; ground snow per the ASCE hazard tool (Louisville criteria) — 30 psf used here (city sheet, non-reducible); Boulder County layer shows 40 psf: engineer confirms |
| **Dead load** | 10 psf deck · 15 psf porch cover (standing seam steel on sheathing, Hardie soffit; 12 psf actual, 15 used) carried on the deck posts |
| **Roof snow (cover)** | 30 psf flat-roof design snow, sliding snow off the house roof + drift at the house wall per ASCE 7 Ch. 7 |
| **Design total** | 50 psf on the deck · cover posts add roof snow + dead to the deck frame and footings (engineered) |
| **Wind** | 145 mph Vult (3-second gust), Exposure C (Louisville 2024 design criteria), Risk Cat. II |
| **Frost depth** | 3'-0" (footings bear below) |
| **Soil bearing** | 1,500 psf presumptive (IRC R401.4.1) — no soils report |
| **Seismic** | B |
| **Wildfire (WUI)** | not in a WUI zone |
| **Codes** | 2024 IRC / IBC as adopted and amended by the City of Louisville (Ord. 1907, Series 2025); IRC R507 decks with LMC 15.05.140; IBC Appendix I patio covers; ASCE 7-22 loads; Louisville Fire Hardening provisions; cold-formed steel frame per AISI S240, engineered |
| **Engineering** | Stamped structural set by a Colorado PE required — the stamped set governs |

## STRUCTURE

| | |
|---|---|
| **Footings** | DP-50/50 |
| **Posts** | 2 x 3x3 galvanized steel post stubs + 5 cover posts carrying the deck beam |
| **Beams** | FLUSH boxed 7-1/4" steel C W: 14'-2", 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16" (worst 474 lb); FLUSH boxed 7-1/4" steel C M: 11'-6", 3 posts at 15'-2" / 20'-8" / 26'-2" (worst 1,616 |
| **Joists** | 7-1/4" CEMCO cold-formed steel C-joists, 16 ga G90, @ 12" OC |
| **Rims** | single-ply 7-1/4" steel track rims and ledger track on a steel angle |
| **Ledger** | 7-1/4" steel ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood |
| **Decking** | TimberTech Landmark American Walnut — 25 rows parallel to the house, picture frame & dividers in Landmark American Walnut, 3/16" gaps, CONCEALoc |
| **Rail** | none |

## SHEET INDEX


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| D-501 | Details |


## SITE

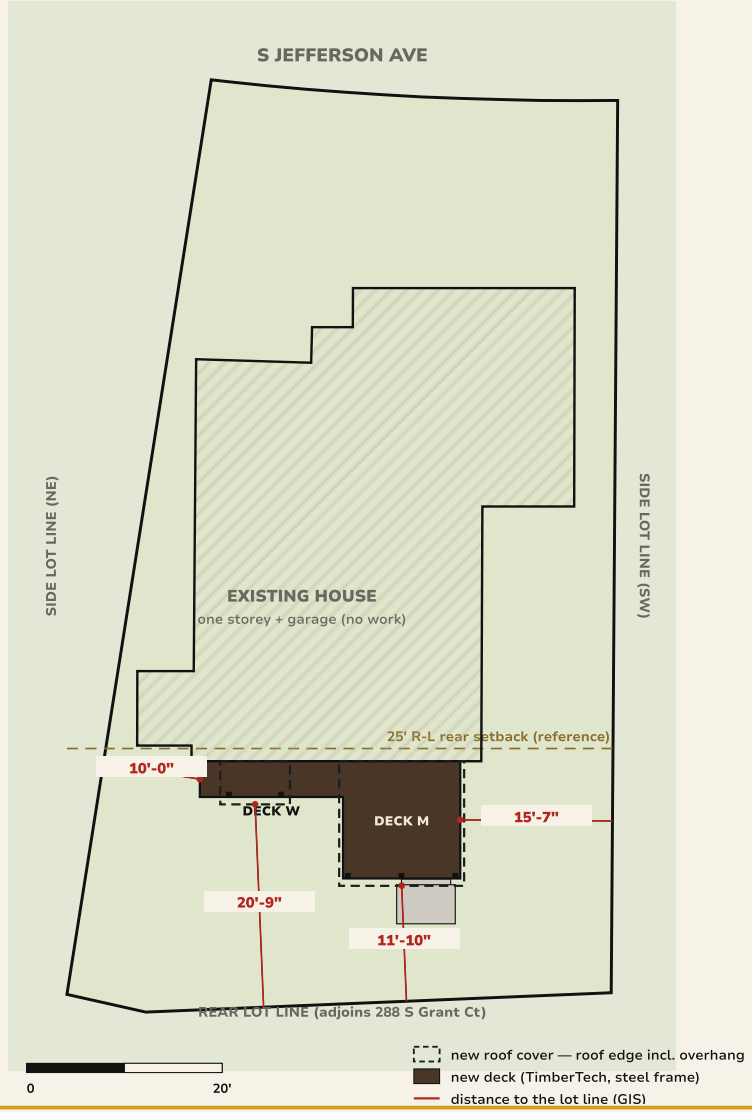
Rear / front lot line: deck edge 12.0' from the line, setback per the PUD...
Side lot line: deck edge 9.8' from the line, setback per the PUD —...

# C-100 SITE PLAN

Parcel, existing house, new decks and roof covers, distances to the lot lines, coverage

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## PARCEL

281 S Jefferson Ave, Louisville CO 80027
Parcel 157517119033 · Account R0092293
Lot 33, Block 10, Dutch Creek (plat rec. 00492001, 1982)
Section 17, T1S, R69W
Zoning: R-L within the Dutch Creek PUD (approved 12/15/1981)
Lot 4,678 SF · flood zone X
House 1982 one-storey ranch: 920 SF main floor + 400 SF garage (assessor)

## LOT COVERAGE

| | |
|---|---|
| **Lot area** | 4,678 SF |
| **House + garage (assessor footprint)** | 1,320 SF |
| **Existing covers to the posts (2024 aerial, est.)** | ~142 SF (main ~12' x 10'-4", walkway ~5'-4" x 3'-5") |
| **Existing coverage** | ~1,462 SF · 31.3% |
| **Main cover, new (to the posts) 12'-0" x 11'-9"** | 141 SF |
| **Walkway cover, new (to the posts) 5'-4" x 3'-5"** | 18 SF |
| **Proposed coverage** | ~1,479 SF · 31.6% (+~17 SF) |
| **Uncovered decks** | not counted (under 30" above grade) |

## SETBACKS

Decks are 30" or less above grade — not held to the principal setbacks (city deck handout)
Roof covers: part of the principal structure — they replace the two existing covers in place; the 2024 aerial shows the existing main roof ~1'-5" shallower: field-verify the existing post line
R-L reference setbacks (LMC 17.12.040) — the Dutch Creek PUD plan governs; Planning to confirm
Measured (red): to the nearest lot line from the new work — GIS, not survey

## SITE WORK

· Remove the two decks, the two roof covers and the box step; haul off
· Diamond Pier foundations (no excavation spoils), 6-mil poly + 2" gravel under the decks
· New concrete landing pad at the stair
· Roof drainage: covers shed to grade at the low eave, away from the house
· Call 811 before the piers; protect the lawn and the owner's blinds

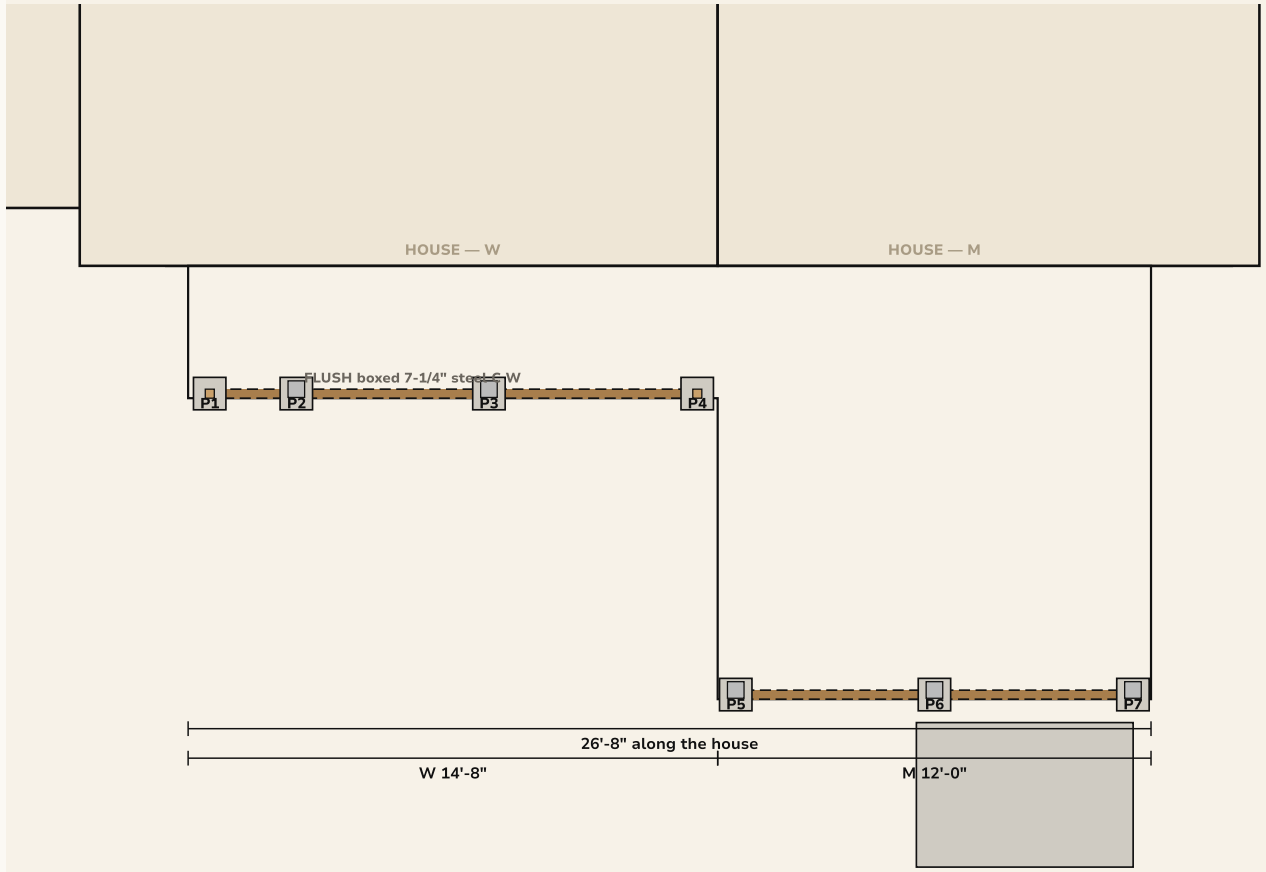
Boulder County GIS parcel + 2024 Pictometry roof outline, Louisville GIS zoning / PUD layers. GIS lines are not a survey — an ILC governs if Planning requires one.

# S-100 FOUNDATION PLAN

Footings, posts and beam lines with the footing schedule; beams dashed (below the joists)

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## FOOTING SCHEDULE

Diamond Pier DP-50/50 (head + 4 x 50" pins), 3,300 lb allowable

| ID | X | Y | ZONE | LOAD |
|---|---|---|---|---|
| P1 | 7 3/16" | 3'-6 1/2" | W | 132 lb |
| P2 | 3'-0" | 3'-6 1/2" | W | 330 lb |
| P3 | 8'-4" | 3'-6 1/2" | W | 474 lb |
| P4 | 14'-1 3/16" | 3'-6 1/2" | W | 273 lb |
| P5 | 15'-2" | 11'-10 1/2" | M | 881 lb |
| P6 | 20'-8" | 11'-10 1/2" | M | 1,616 lb |
| P7 | 26'-2" | 11'-10 1/2" | M | 881 lb |

x from the left frame face facing the house; y from the reference wall.
Frost 3'-0" · soil 1,500 psf · worst post 1,616 lb

Plan · not to scale · house at the top · dimensions govern

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

# S-101 FRAMING PLAN

Ledger, joists, rims, blocking, beams; every member in the model — the cut list is on T-600

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## FRAMING NOTES

7-1/4" CEMCO cold-formed steel C-joists, 16 ga G90, @ 12" OC

single-ply 7-1/4" steel track rims and ledger track on a steel angle

7-1/4" steel ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board

blocking one row over the beam, staggered

FLUSH boxed 7-1/4" steel C W: 14'-2", 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16" (worst 474 lb); FLUSH boxed 7-1/4"

Connectors:

Steel clip angle 1-1/2" x 1-1/2" x 7" 16 ga G90 — joist to track / beam: 56

Lateral-load tension tie for steel joists (Simpson DTT2Z w/ steel-joist kit or the engineer's detail) — IRC R507.9.2

Cold-galvanizing spray (cut ends, field drilling): 2



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# A-101 DECKING PLAN

Every board, border and divider as it lays; rail posts shown for the pockets

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## DECKING

Field: TimberTech Landmark American Walnut, 5.5" x 1", 3/16" gaps

Border & dividers: Landmark American Walnut

Boards run parallel to the house; edge overhang 1" over the fascia

CONCEALoc at every joist in every gap (437)

Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then

Rows per panel: panel 1 (0"–14'-8") 7 rows · panel 2 (14'-8"–26'-8") 25 rows · 2 ripped pieces · 1 breaker — every board one piece wall to border, no butt

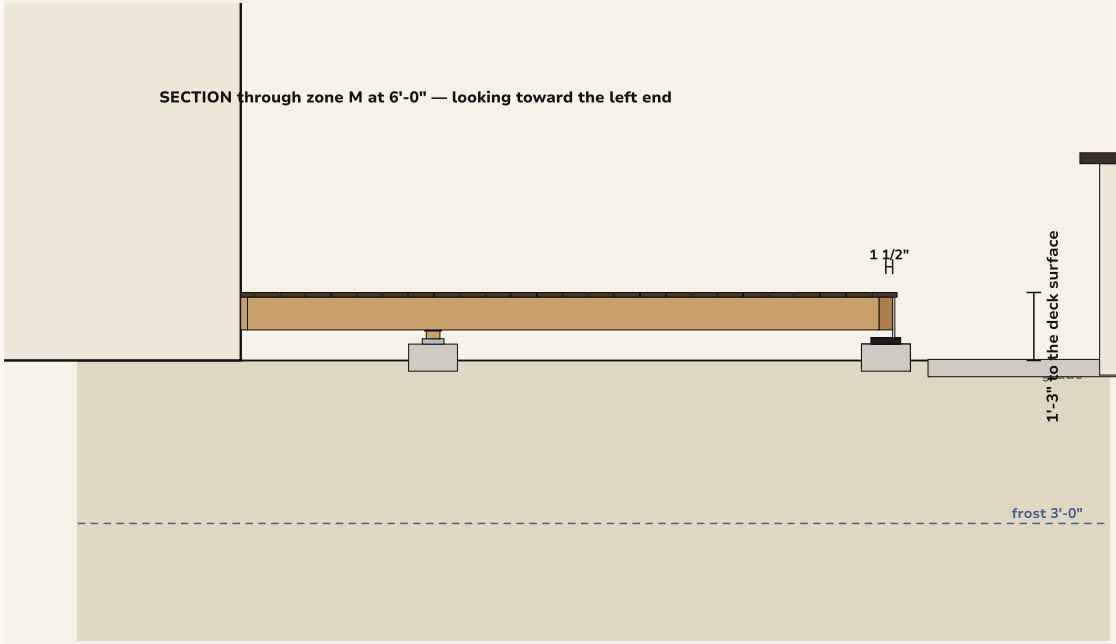
# A-301 SECTION + FRONT ELEVATION

Cut through the deepest zone; heights to grade, frost line, cantilever, guard

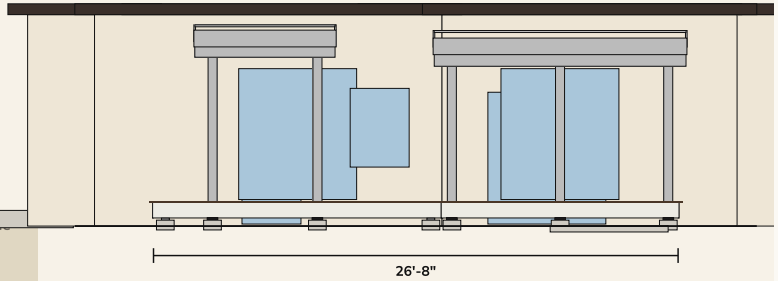
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**SECTION through zone M at 6'-0" — looking toward the left end**




**FRONT ELEVATION — from the yard**




## SECTION NOTES

Deck surface 1'-3" above grade

FLUSH boxed 7-1/4" steel C W: 14'-2", 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16" (worst 474 lb); FLUSH boxed 7-1/4" steel C M: 11'-6", 3 posts at 15'-2" / 20'-8" / 26'-2" (worst

Posts 2 x 3x3 galvanized steel post stubs + 5 cover posts carrying the deck beam on DP-50/50

7-1/4" steel ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood

Section left · elevation right · not to a common scale

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

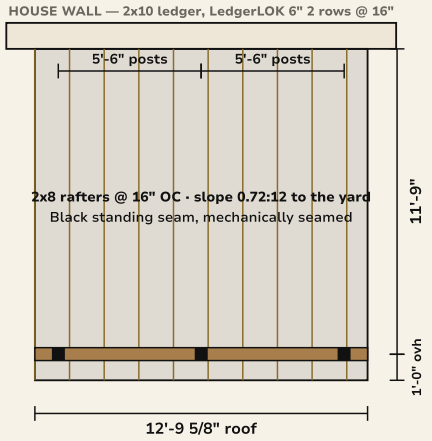
# A-401 ROOF COVERS — PLAN + SECTION

Lean-to covers: ledger, rafters, beam, posts on the piers, clear heights, pitch, standing seam, Hardie, blinds

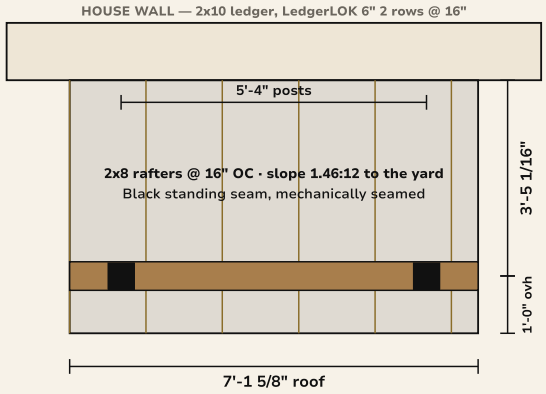
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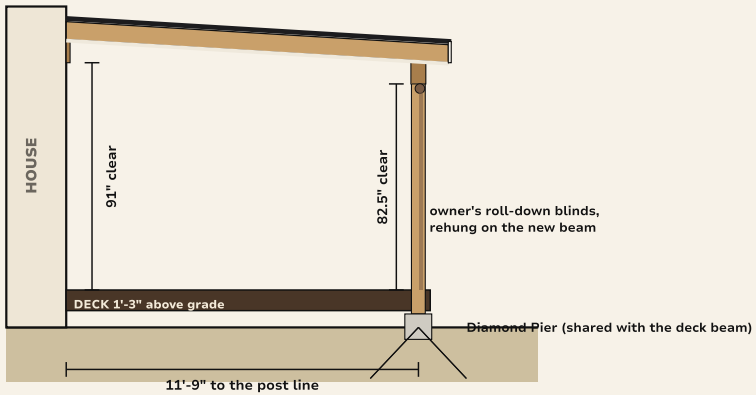
## MAIN COVER — PLAN




## WALKWAY COVER — PLAN

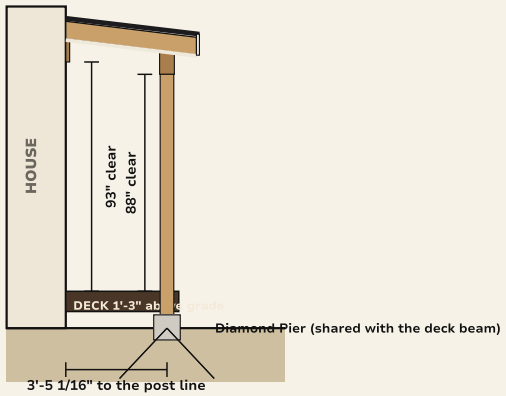



## MAIN COVER — SECTION




Standing seam 24 ga, 16" panels, 1-1/2" mechanical (double-lock) seam — rated for 0.72:12 · high-temp ice & water underlayment over 7/16" OSB
Headwall flashing counterflashed into the siding · color-matched eave / rake trim · snow bar across the low eave
HardieSoffit cedarmill ceiling · HardieTrim fascia + rakes (Arctic White) · H2.5AZ at every rafter · AC6Z caps
6x6 posts on ABA66Z bases on the Diamond Pier bolt · the engineer's uplift hold-down governs

## WALKWAY COVER — SECTION




Standing seam 24 ga, 16" panels, 1-1/2" mechanical (double-lock) seam — rated for 1.46:12 · high-temp ice & water underlayment over 7/16" OSB
Headwall flashing counterflashed into the siding · color-matched eave / rake trim · snow bar across the low eave
HardieSoffit cedarmill ceiling · HardieTrim fascia + rakes (Arctic White) · H2.5AZ at every rafter · AC6Z caps
6x6 posts on ABA66Z bases on the Diamond Pier bolt · the engineer's uplift hold-down governs

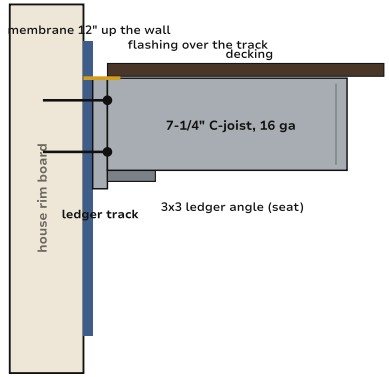
# D-501 DETAILS — STEEL FRAME

Steel ledger · joist clips · boxed beam on the shared pier · Diamond Pier · deck edge · under the deck

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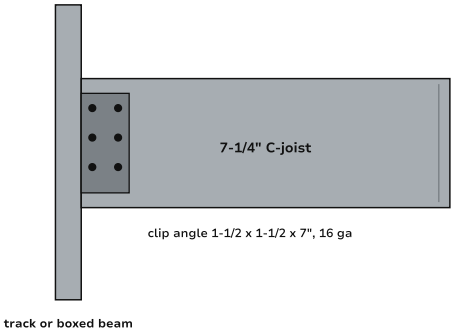
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## 1 STEEL LEDGER AT THE HOUSE



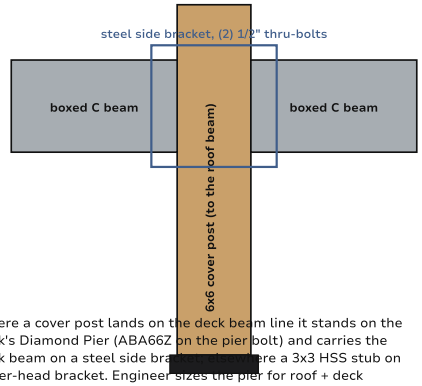

7-1/4" steel ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood

## 2 JOIST TO TRACK / BEAM



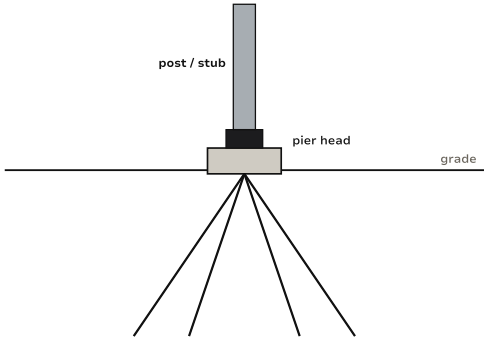

#12-14 self-drilling hex-washer screws, 4 per leg minimum (clip to joist web, clip to track / beam web) — the engineer's clip schedule governs. Cut ends and field holes cold-galvanized.

## 3 BOXED BEAM · SHARED PIER



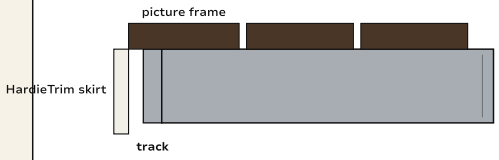

Where a cover post lands on the deck beam line it stands on the deck's Diamond Pier (ABA66Z on the pier bolt) and carries the deck beam on a steel side bra[illegible]re a 3x3 HSS stub on a pier-head bracket. Engineer sizes the pier for roof + deck and the uplift hold-down.

## 4 DIAMOND PIER




DP-50/50 (4 pins) — no excavation; frost-protected by the pins (submit the evaluation report). 811 locate first.

## 5 DECK EDGE · HARDIE SKIRT




TimberTech Landmark American Walnut: hidden fasteners in the field, the picture frame face-fastened with color-matched plugs — every screw the maker's steel-frame screw for 14–18 ga. HardieTrim skirt on furring screwed to the track.

## 6 UNDER THE DECK · LOW END



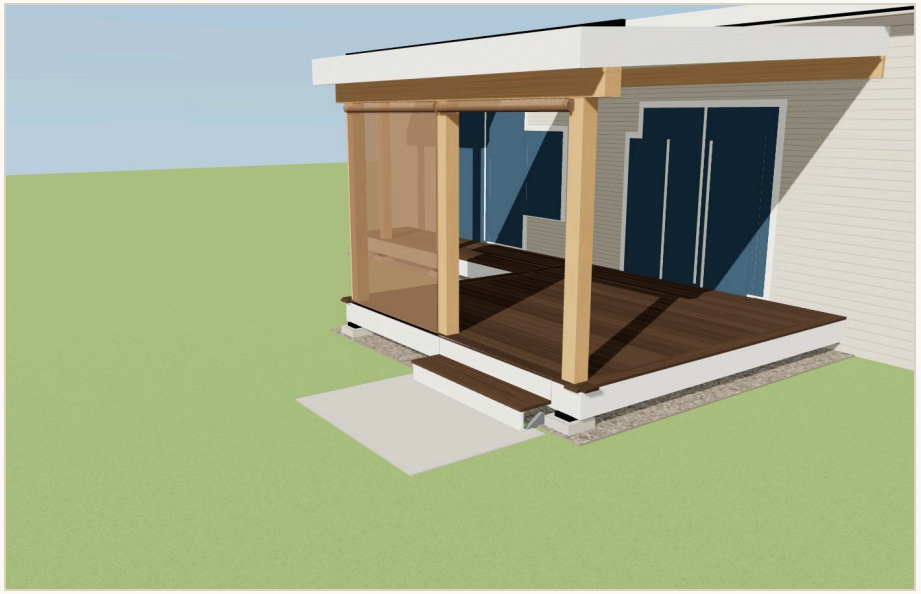

2" gravel over 6-mil poly, sloped away from the house

Walkway low point: 9" surface to grade leaves 0.75" under the frame — dig out / regrade to 2" min. under that run before the poly.

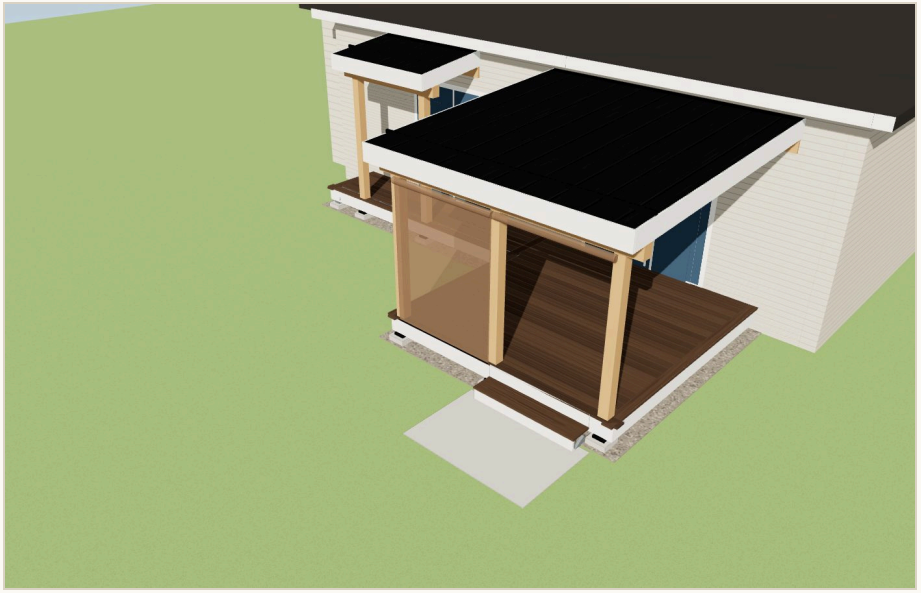
## Decks + roof covers



From the yard



From the front corner



Isometric



Under the deck